

QorFinder

Presentation

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Introduction

Why QorFinder?



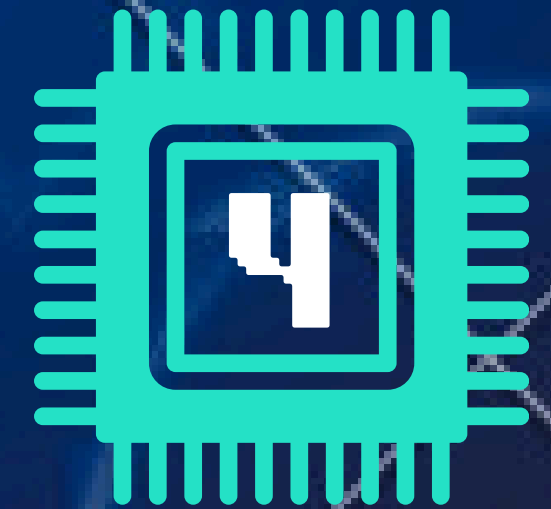
Literature Review

Find out the competitors
and approaches available.



Project Scope

Where should
QorFinder start?



Methods Involved

How to create
QorFinder?



Introduction

Why QorFinder?

Problems encountered

Files pile up quickly

- accumulate thousands of files and reference materials
- which leads to an overwhelming state of digital clutter

Forgot the file name

- recall the meaning instead of the file name
- does not remember the file name of an invoice issued last week, 234567890dhvxfhn.pdf

Costly to organize

- batch renaming tools available, however requires extra effort to manage
- prone to changes, the organized file name lost its meaning when the usage changes

Why we need to solve it?

A new report from Atlassian has highlighted significant inefficiencies faced by Australian knowledge workers, revealing that they spend a considerable 23.5% of their work week, roughly 10 hours, searching for information.

The findings from the report cast a light on the time lost by knowledge workers as they scour through emails, messaging channels, and disparate systems to find the data they need. A total of 73% of these workers reported challenges in locating the necessary information to accomplish their tasks, pointing to widespread productivity issues driven by information barriers.

Why Not RAG?

Here are some reasons...

It is too heavy for an average computer to run a local LLM.

In some scenarios, users don't want AI summaries.

Not willing to rely on cloud services as it implies extra expenses

Who will benefit from QorFinder?

ordinary
Knowledge Workers

and also specialised



Technical Professionals

Software developers
navigating in tons of
documentation and codes.

Privacy-Conscious Practitioners

Legal professionals and
medical providers
handling sensitive client
data.

SME Business Owners

Budget-constrained users
requiring functionality without
any additional costs.

Motivation

It happen on me.

Lack of money to purchase hardware

Care about my privacy

Wants absolute control on my data



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Literature Review

Find out the competitors
and approaches available.

Previous Works

Previous works on File Retrieval

Lexical File Search

- Voidtools' "Everything"
- ripgrep

Semantic File Search

- FileBrain
- MangoFinder

Batch File Rename Tools

- Bulk Rename Utility
- Renamer.ai

Previous works on Sentence Embeddings

BERT

SBERT

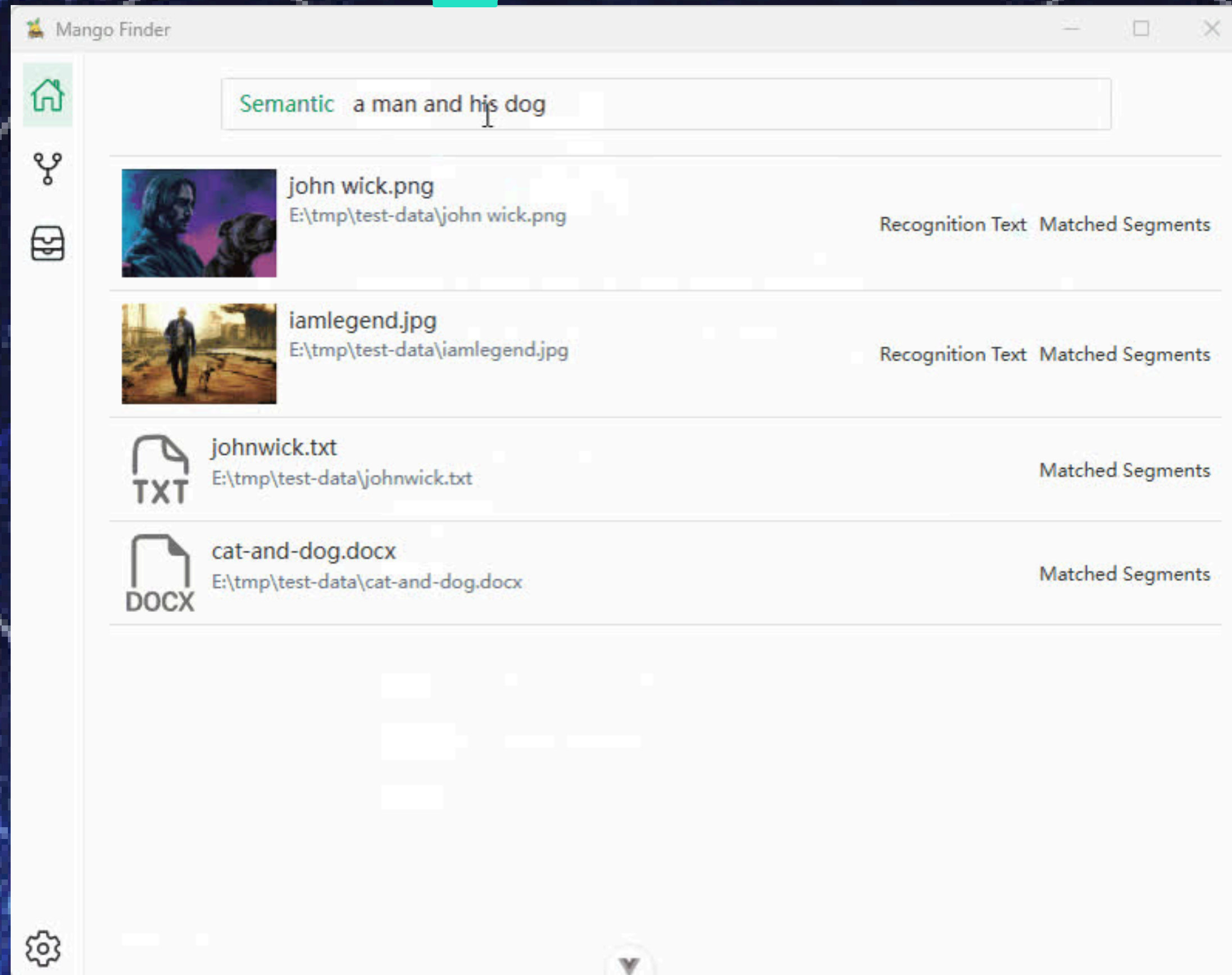
- SBERT-WK
- TA-SBERT

Bi-Encoders

Cross-Encoders

MangoFinder

ONNX
SQLite
Rust



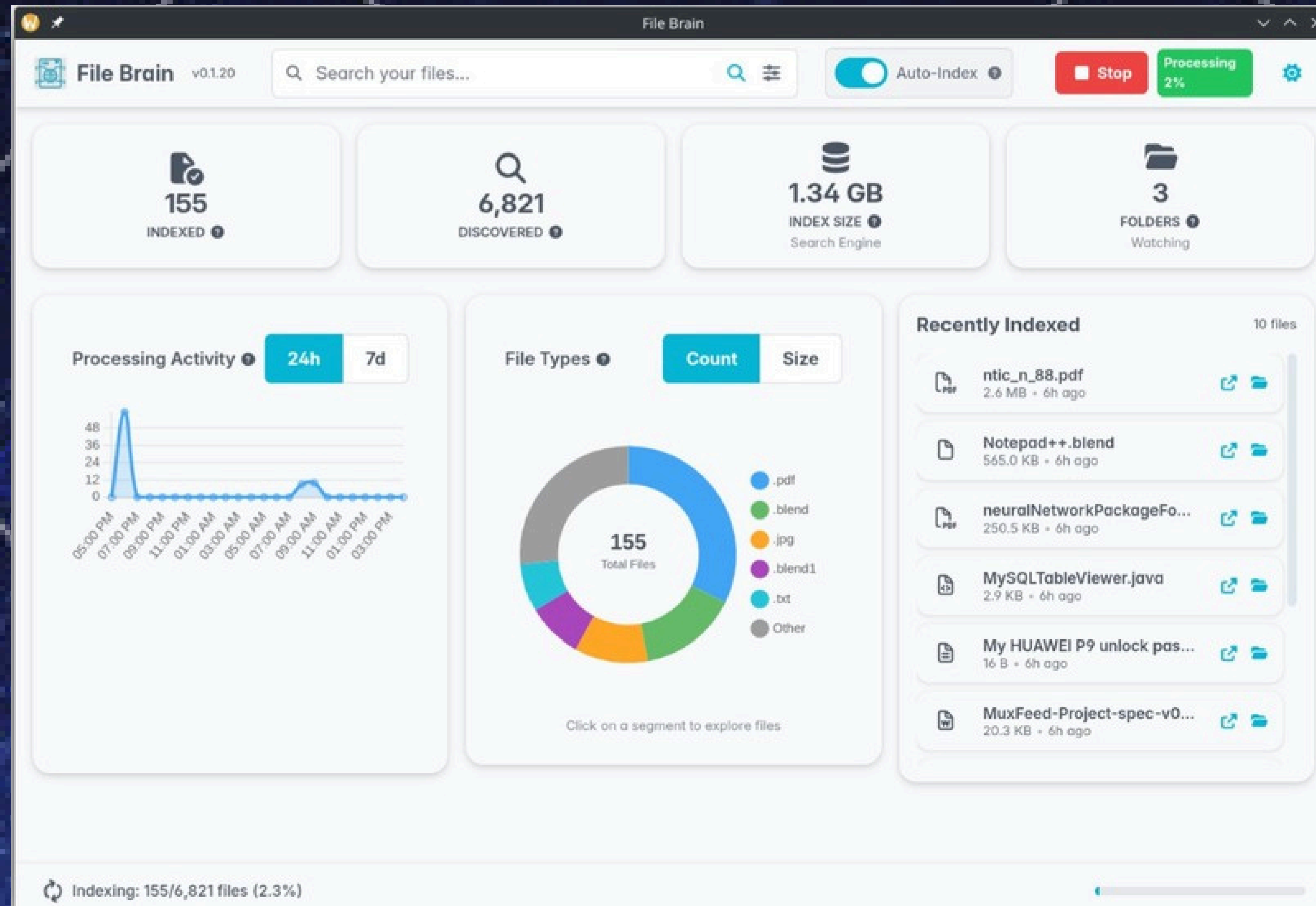
MangoFinder

Features:

- Cross-device search
- Multilingual & Cross-language Search
- Integration with Ollama
- Supports text, image and audio
- Cross-platform compatibility

File Brain

Apache Tika
TypeSense
Python
Docker

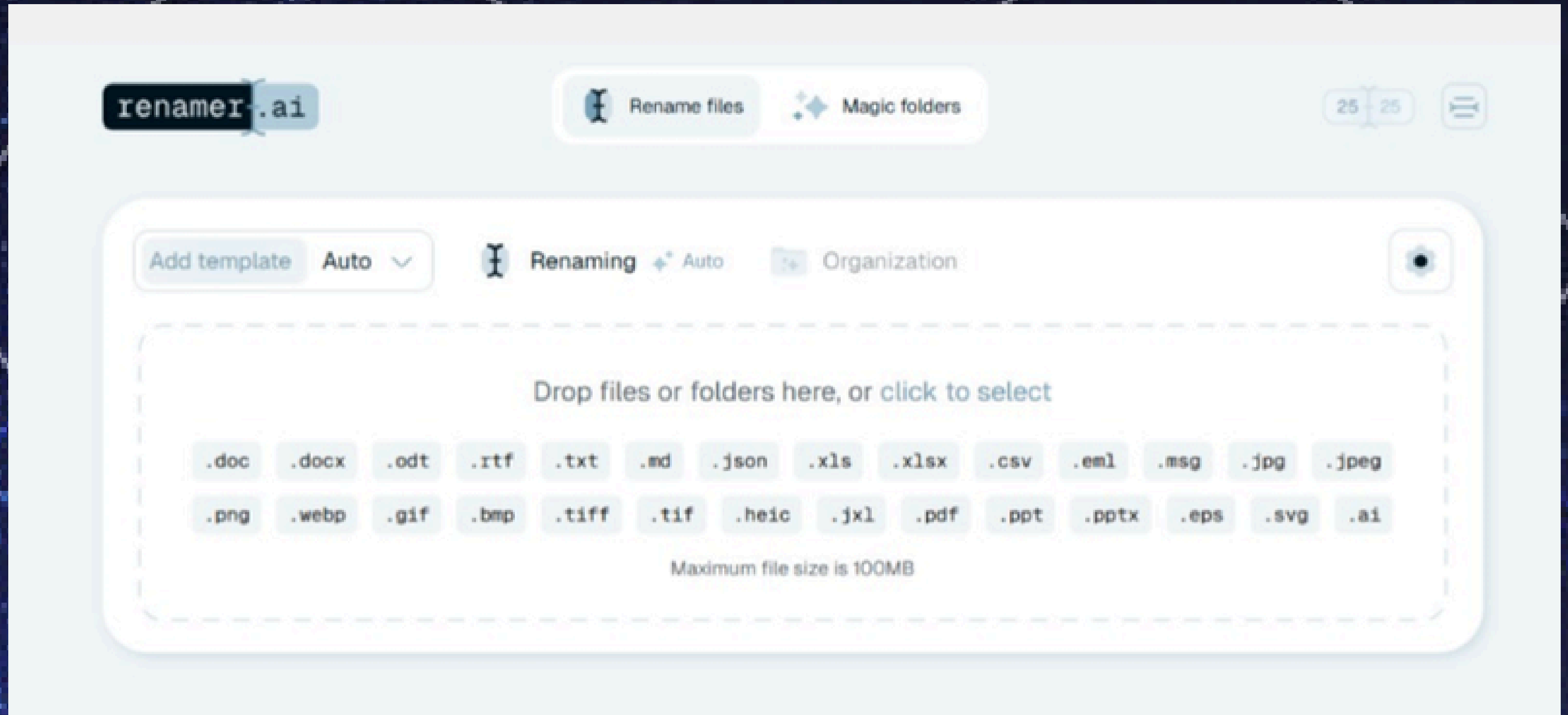


File Brain

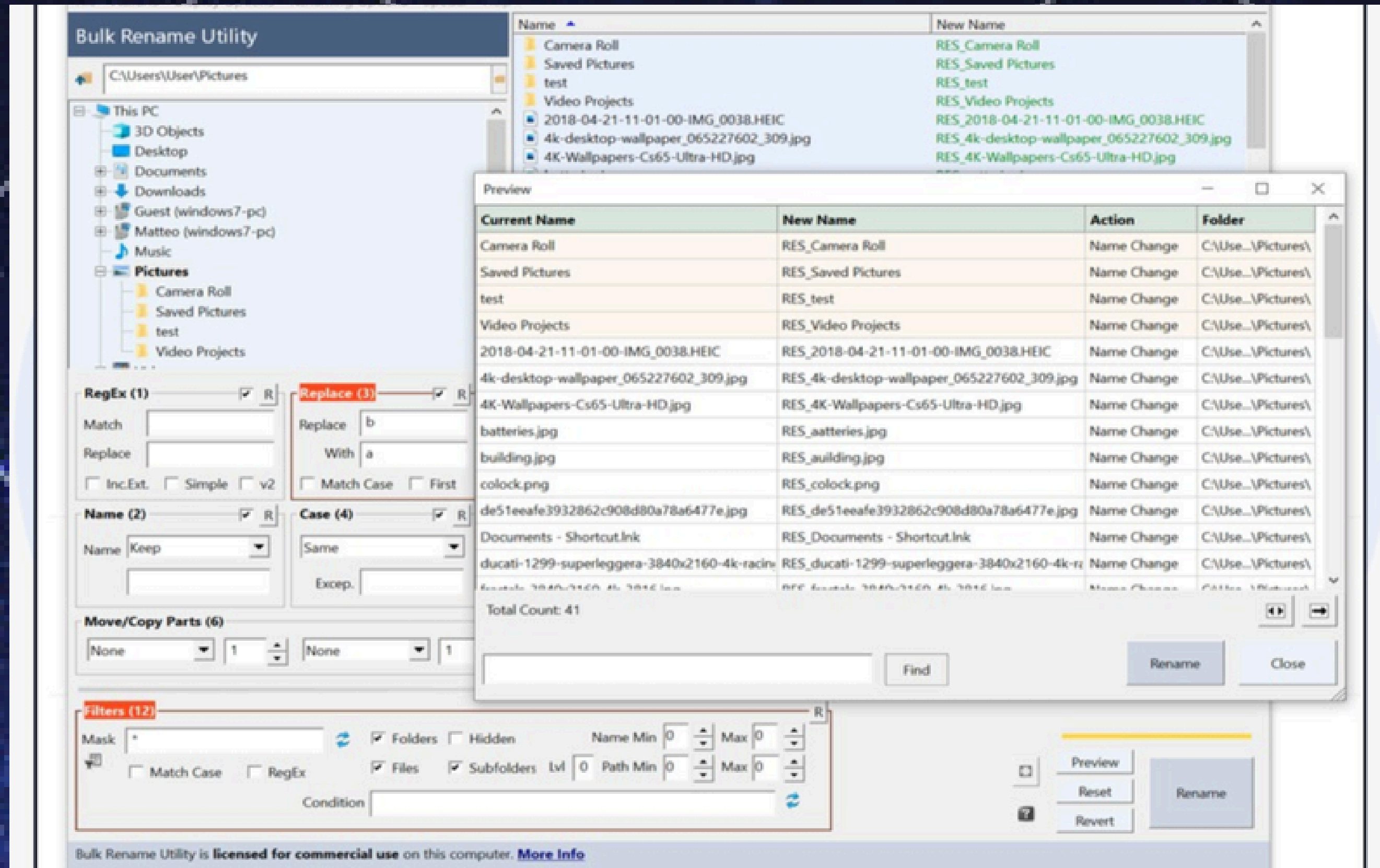
Features:

- In-memory search engine (powered by Typesense)
- Content Extraction (powered by Apache Tika)

Renamer.ai



Bulk Rename Utility



Everything

 *w9*.pdf - Everything

File Edit View Search Bookmarks Tools Help

w9.pdf

Name	Path	Size	Date Modified
 fw9.pdf	G:\Archives\Informa	280 KB	3/9/2020 5:08 PM
 fw9-2020-signed.pdf	C:\Users\etitt\Documents	244 KB	1/5/2020 5:58 PM
 fw9-2020-signed (2020_12_18 17_08_50 UTC).pdf	K:\FileHistory\etitt\J7SKYLAKE\Data\C\Users\etitt\Documents	244 KB	1/5/2020 5:58 PM
 fw9-2020.pdf	C:\Users\etitt\Documents	130 KB	1/5/2020 5:56 PM
 fw9-2020 (2020_12_18 17_08_50 UTC).pdf	K:\FileHistory\etitt\J7SKYLAKE\Data\C\Users\etitt\Documents	130 KB	1/5/2020 5:56 PM
 190106-W9-signed.pdf	F:\ETPersonal\ETDocs	267 KB	1/6/2019 11:08 AM
 Oct18w9.pdf	G:\Archives\Pictures	130 KB	1/6/2019 11:05 AM
 Dec2017-w9-signed.pdf	G:\Archives\Documents	293 KB	12/21/2017 1:15 PM
 2017-w9-print.pdf	G:\Archives\Documents	270 KB	9/20/2017 11:24 AM
 2017-w9.pdf	G:\Archives\Documents	117 KB	8/16/2017 4:36 PM
 fw9-signed.pdf	F:\ETPersonal\Ciena	168 KB	11/22/2013 1:44 PM
 fw9.pdf	F:\ETPersonal\Ciena	71 KB	11/22/2013 1:41 PM
 ert-w9-2012.pdf	F:\ETPersonal\10Taxes	331 KB	4/17/2012 5:57 PM
 ert-w9-2010.pdf	F:\ETPersonal\10Taxes	245 KB	12/14/2010 2:47 PM

< >

14 objects

Everything

NTFS Master File Table (MFT) Access: Bypasses traditional directory traversal by reading the MFT directly to create an instant map of the drive.

Global Efficiency: Widely recognized as the most efficient engine for large-scale file searching on Windows.

Instantaneous Performance: Capable of processing searches across millions of files in real-time.

RipGrep

```
[andrew@Cheetah rust] rg -i rustacean
src/doc/book/nightly-rust.md
92:[Mibbit][mibbit]. Click that link, and you'll be chatting with other Rustaceans

src/doc/book/glossary.md
3:Not every Rustacean has a background in systems programming, nor in computer

src/doc/book/getting-started.md
176:Rustaceans (a silly nickname we call ourselves) who can help us out. Other great
376:Cargo is Rust's build system and package manager, and Rustaceans use Cargo to

src/doc/book/guessing-game.md
444:it really easy to re-use libraries, and so Rustaceans tend to write smaller

CONTRIBUTING.md
322:* [rustaceans.org][ro] is helpful, but mostly dedicated to IRC
333:[ro]: http://www.rustaceans.org/
[andrew@Cheetah rust] □
```

RipGrep

Tool	Command	Line count	Time
ripgrep (Unicode)	<code>rg -n -w '[A-Z]+_SUSPEND'</code>	536	0.082s (1.00x)
hypergrep	<code>hgrep -n -w '[A-Z]+_SUSPEND'</code>	536	0.167s (2.04x)
git grep	<code>git grep -P -n -w '[A-Z]+_SUSPEND'</code>	536	0.273s (3.34x)
The Silver Searcher	<code>ag -w '[A-Z]+_SUSPEND'</code>	534	0.443s (5.43x)
ugrep	<code>ugrep -r --ignore-files --no-hidden -I -w '[A-Z]+_SUSPEND'</code>	536	0.639s (7.82x)
git grep	<code>LC_ALL=C git grep -E -n -w '[A-Z]+_SUSPEND'</code>	536	0.727s (8.91x)
git grep (Unicode)	<code>LC_ALL=en_US.UTF-8 git grep -E -n -w '[A-Z]+_SUSPEND'</code>	536	2.670s (32.70x)
ack	<code>ack -w '[A-Z]+_SUSPEND'</code>	2677	2.935s (35.94x)

RipGrep

- Rust-Powered
- Optimized Regex engine
- Linear Time Search: Uses the Aho-Corasick algorithm to search for multiple strings simultaneously without slowing down as the query grows.
- SIMD Optimizations: Leverages Single Instruction, Multiple Data to process large blocks of text in parallel at the CPU register level.
- "Teddy" Algorithm: Utilizes a specialized literal matcher (borrowed from Intel's Hyperscan) for ultra-fast scanning.

The Transformer Paradigm

Parallel Token Processing

Context-aware relationship captured
via self-attention mechanism

Core bases of BERT (Bidirectional
Encoder Representations)

Utilises masked language modelling for
high-fidelity semantic accuracy

Inference Bottleneck

65h

Inference Time (BERT)

While BERT achieves high accuracy, its architectural requirement for joint pair processing limits real-world application. Comparing 10,000 sentences requires massive computation, resulting in a 65-hour latency that prohibits real-time local deployment.

Siamese Network



Inference Time (SBERT)

The SBERT Innovation

Sentence-BERT (SBERT) utilizes twin subnetworks with shared weights to map disparate inputs into a unified vector space.

By applying mean pooling to aggregate token-level data into fixed-size embeddings, it enables ultra-fast Cosine Similarity searches.

Advanced Pooling Variants

ordinary

SBERT

derived some variants

SBERT-WK

Uses geometric layer fusion for a robust 'semantic fingerprint', considering syntactic components often omitted in baseline pooling.

TA-SBERT

Implements Token-Level Attention to assign weights based on significance, ensuring critical tokens dominate the sequence vector.

Performance

Geometric and attention methods yield 10—15% gains over standard mean pooling benchmarks in STS tasks.

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Project Scope

Where should QorFinder start?

Objective

To design, engineer, and deploy an efficient and private semantic file search desktop application that supports the fast and effective searching of English documents via natural language inputs using only local hardware without cloud services.

—To design and implement a loosely-coupled architecture

—To iterate and fine-tune the sentence transformer models

—To test results through performance benchmarking

Scope

On Windows 10/11

Built with Rust and Tauri

High-speed semantic
search (< 16GB RAM)

Exclusively indexes
text-based documents:
.txt, .pdf, .docx, and
.md.

Powered by Qdrant (vector
database) and Sentence
Transformers for offline
indexing.

Entirely offline operation

Most importantly, now it will be focusing on English only, possibly include Malay if time given is sufficient

Data Flow Diagram

— There are 2 proposed workflows, where how QorFinder should work

1

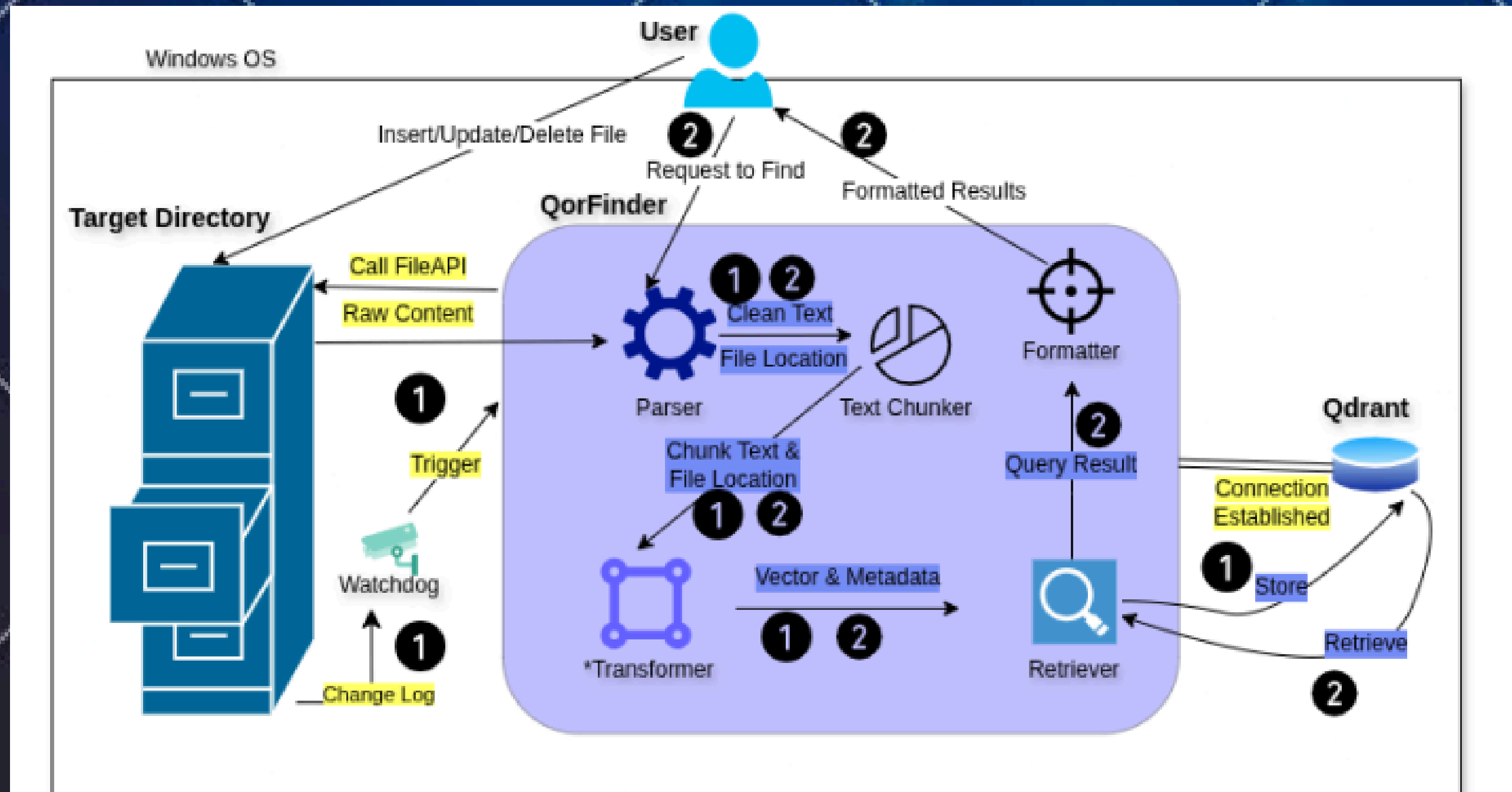
Initialisation
or
Background
Daemon

2

Semantic
Search

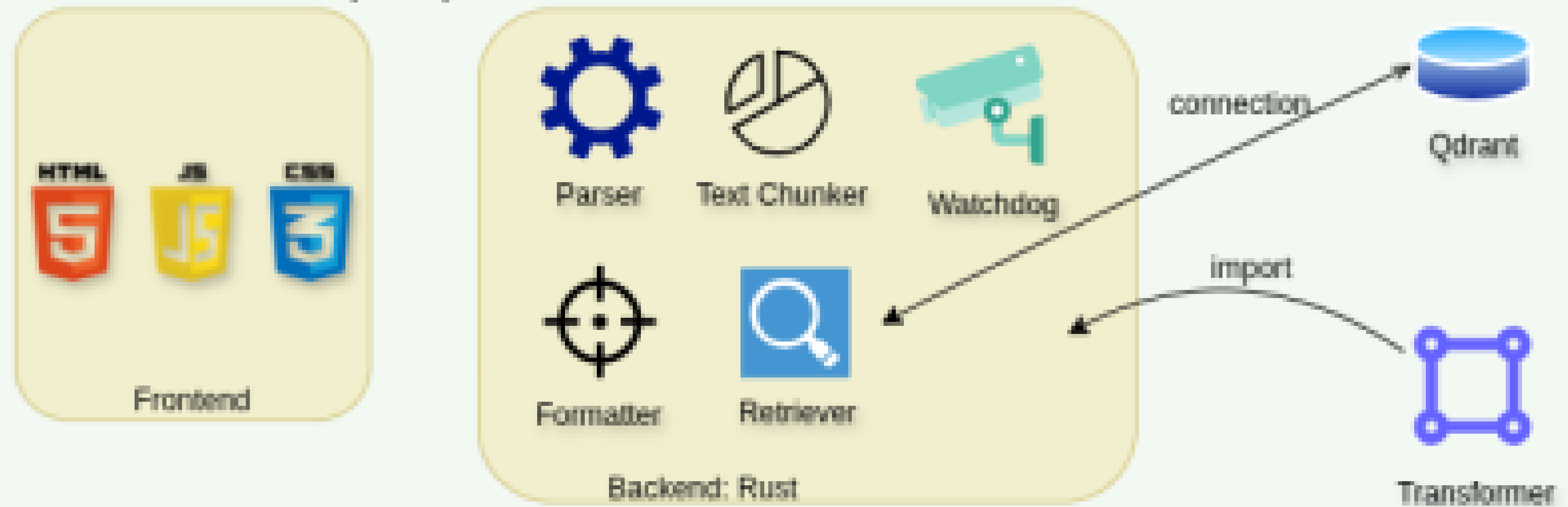
Data Flow Diagram

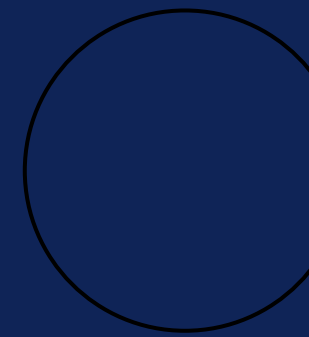
— There are 2 proposed workflows, where how QorFinder should work



QorFinder

Framework: Tauri (Rust)

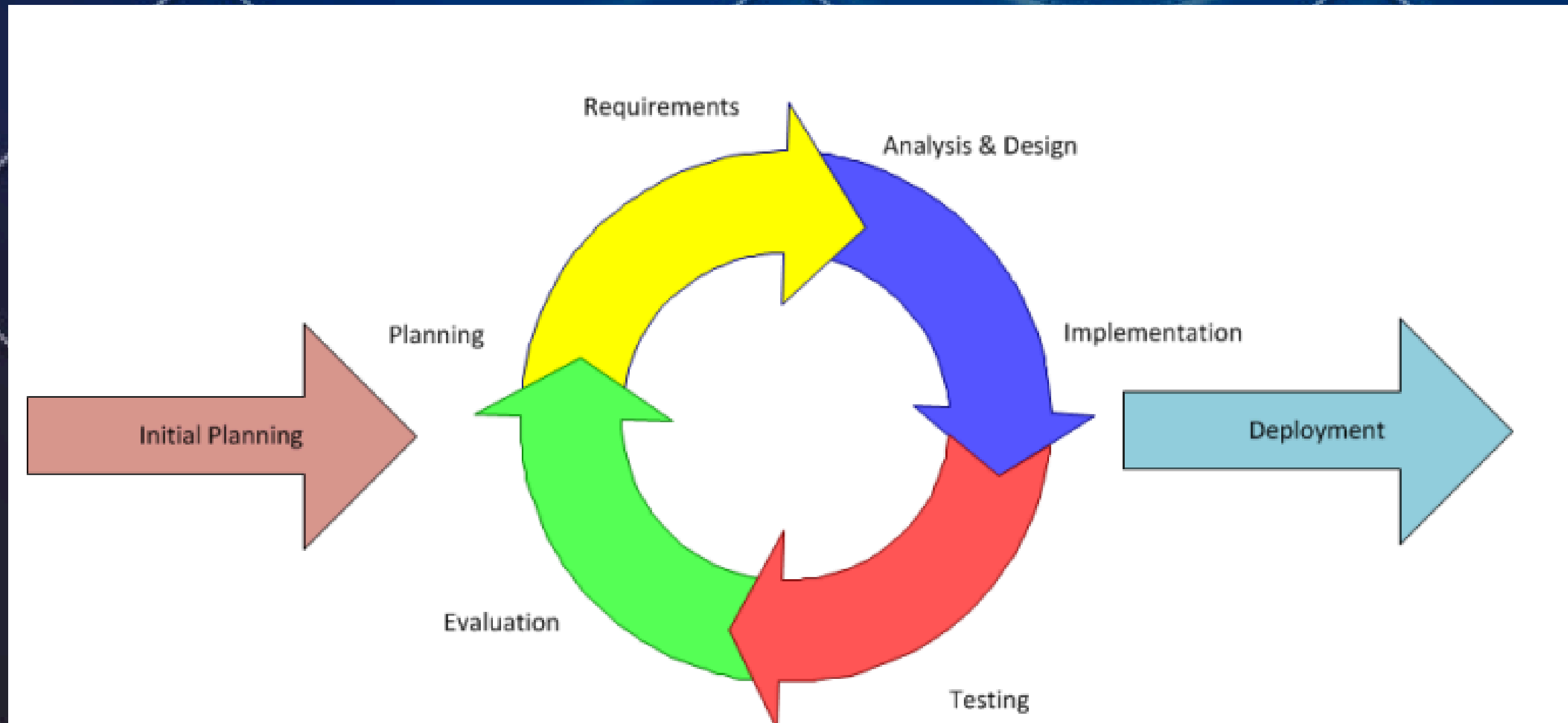




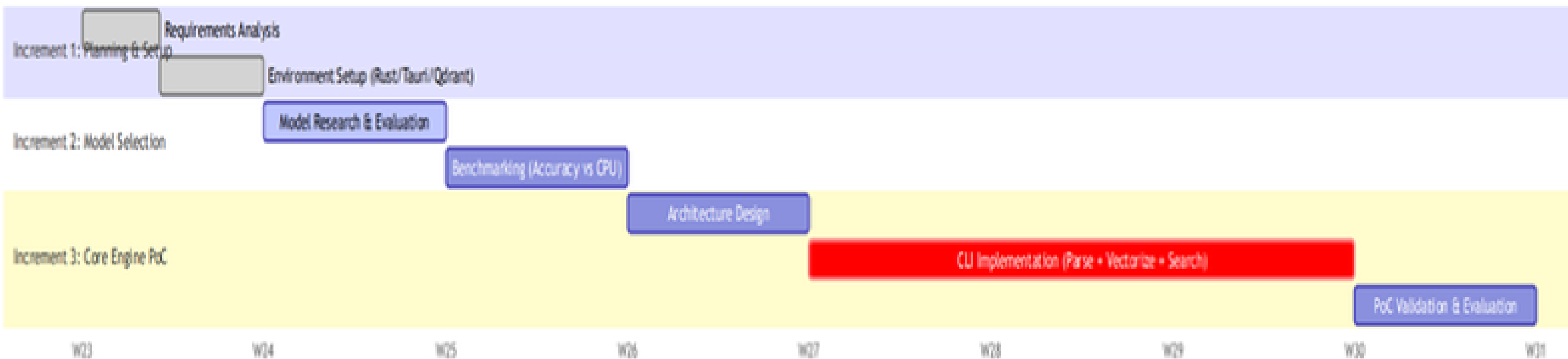
Methods Involved

How to create QorFinder?

Iterative and Incremental Development (IID)



FYP1



FYP2



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Thank you!

Appreciate to Dr. Hafizul's attention.

Resource Page

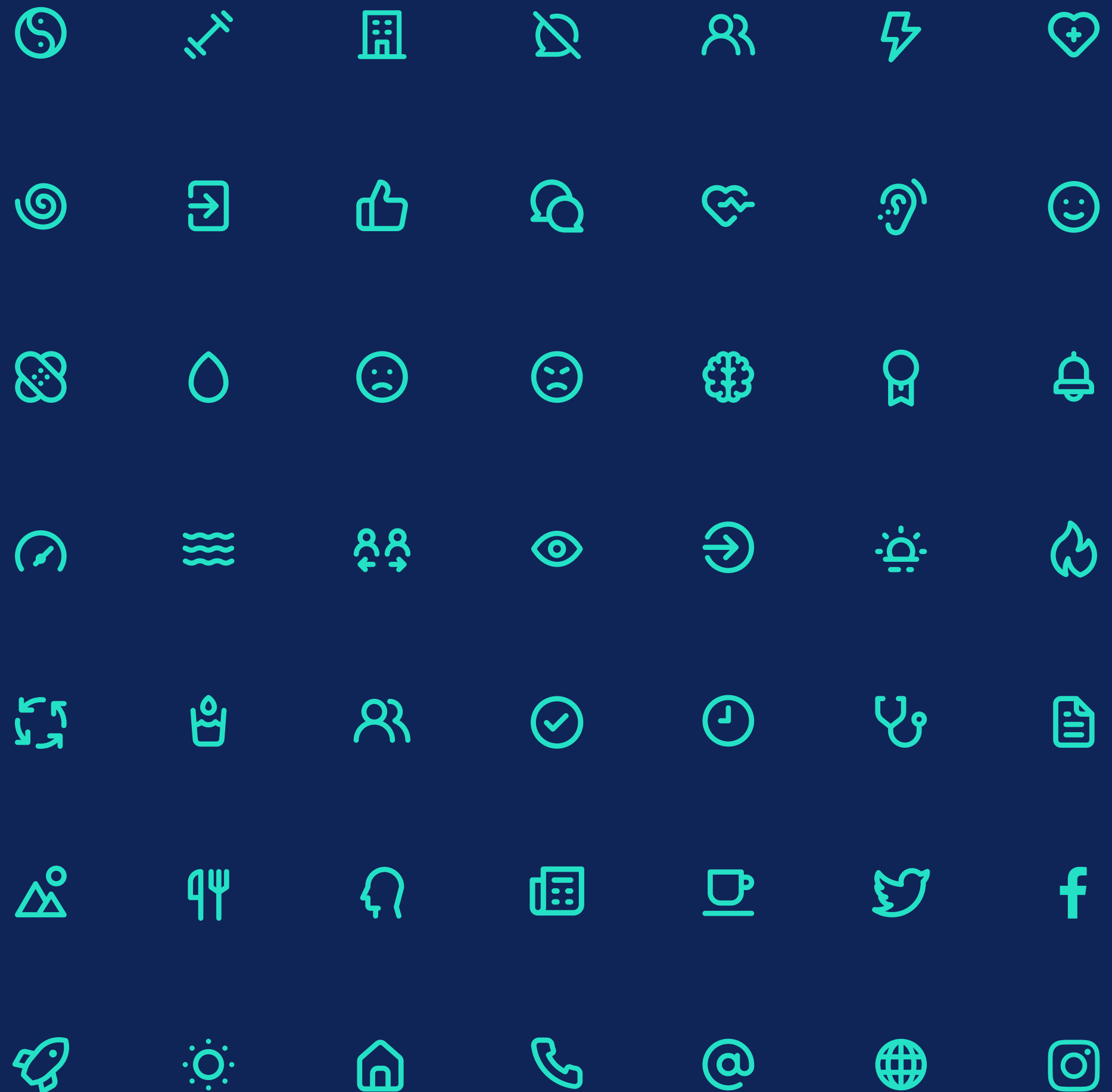
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